

Program : Diploma in Textile Technology	
Course Code : 5067	Course Title: Textile Testing I Lab
Semester : 5	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

1. To provide thorough knowledge of determination of RH, moisture regain, and Analysis of structure of Textile fibres.
2. To equip students with the skill of determining fibre properties such as fibre length, fineness, and maturity.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of Textile Fibres		Textile Fibres	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Execute the testing for atmospheric conditions & Regain of textile materials.	8	Applying
CO2	Interpret cotton fiber length and fineness by suitable instruments.	14	Applying
CO3	Demonstrate strength, maturity, and trash content in cotton fiber.	11	Applying
CO4	Study of cotton fibre properties tests by AFIS and HVI	9	Understanding
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Execute the testing for atmospheric conditions & Regain of textile materials.		
M1.01	Determine Relative humidity of atmosphere by wet and dry bulb and sling hygrometer	3	Applying
M1.02	Estimation of moisture content and regain by conditioning oven	3	Applying
M1.03	Study the method of determining moisture regain by electronic moisture meter	2	Applying
CO2	Interpret cotton fiber length and fineness by suitable instruments.		
M2.01	Identify the important textile fibers using microscopic examination	4	Applying
M2.02	Determination of Cotton fiber length using bear sorter	4	Applying
M2.03	Study of Cotton fiber length determination using digital fibro graph	3	Applying
M2.04	Determination of Cotton fiber fineness by fiber fineness tester	3	Applying
	Lab Exam - I	1.5	
CO3	Demonstrate strength, maturity, and trash content in cotton fiber.		
M3.01	Determination of cotton Fiber bundle strength by Stelometer.	4	Applying
M3.02	Determination of Cotton fiber maturity by caustic soda swelling method	4	Applying
M3.03	Estimate trash content in cotton by trash analyzer	3	Applying

CO4	Demonstrate the dyeing of cotton/ viscose material using water-insoluble dyes and analyse the fastness properties.		
M4.01	Study of Cotton fiber properties testing by AFIS	5	Understanding
M4.02	Study of Cotton fiber properties testing by HVI	4	Understanding
	Lab Exam - II	1.5	

The students should keep record for wet processing with samples and submit certified & bonafide record at the time of practical examination.

Text / Reference:

T/R	Book Title/Author
T1	J.E. BOOTH - Principles of Textile Testing (Butterworth third edition)
R2	Grover - Hand book of Quality control and Textile testing Butterworth
R3	SITRA - Principles of Textile Testing
R4	ISO - Hand book of textile testing

Online Resources:

Sl.No	Website Link
1	https://www.astm.org/Standards/textile-standards.html
2	https://www.testertextile.com
3	www.uster.com/en/instruments/fiber-testing/uster-afis-pro/
4	www.youtube.com